

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Computer Science and Engineering

### ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

|                  |                                                                                                                                                                |               |                                                 |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------------------|
| Course Code:     | CSA06                                                                                                                                                          | Course Title: | Design and Analysis of Algorithms               |
| CO Assessed:     | CO1 – Precision (BL1, BL2, BL3)                                                                                                                                | Assessment:   | Assessment Tool 1 – Application Based Questions |
| Weightage:       | 40% of CIA                                                                                                                                                     | Total Marks:  | 100 (2 Questions × 50 Marks)                    |
| CO1 Statement:   | <i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i> |               |                                                 |
| Student Name:    | Y. Greeshma Reddy                                                                                                                                              | Reg. No.:     | 192424146                                       |
| Section / Batch: | B. Tech-AI&DS                                                                                                                                                  | Date:         | 16-07-2026                                      |

#### Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

| Question Type               | Focus Area                                                           | Questions | Marks              |
|-----------------------------|----------------------------------------------------------------------|-----------|--------------------|
| Application-Based Questions | Apply Master Theorem and asymptotic analysis to real-world scenarios | Q1 – Q2   | Q1 – 50<br>Q2 – 50 |
| GRAND TOTAL:                |                                                                      | 100 Marks |                    |

#### SECTION A – APPLICATION BASED QUESTIONS

**Q1.** Cybersecurity Threat Analysis System A cybersecurity system analyzes threats using the recurrence  $T(n)=T(n-1)+n\log n$ . Apply the substitution method to solve the recurrence. Expand step-by-step and derive the complexity. Analyze how the algorithm scales with increasing threat data. Interpret efficiency using asymptotic notation.

**Q2.** E-Commerce Recommendation System An e-commerce platform processes product recommendations using the recurrence:  $T(n) = 3T(n/3) + n$ . Apply the Master Theorem to determine the time complexity. Clearly identify parameters and justify the selected case. Analyze how this affects system performance as the number of users grows. Interpret efficiency using asymptotic notation.

**Code of Conduct**

I Y. Greeshma Reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Y. Greeshma Reddy

**RUBRICS FOR CALCULATION**

| Criteria                         | Wt. | Level 4 – Exemplary                                                        | Level 3 – Proficient                                  | Level 2 – Developing                           | Level 1 – Beginning                                    |
|----------------------------------|-----|----------------------------------------------------------------------------|-------------------------------------------------------|------------------------------------------------|--------------------------------------------------------|
| Understanding of Problem Context | 20% | Clearly understands the problem and accurately identifies all requirements | Good understanding with minor gaps                    | Basic understanding; some requirements unclear | Poor understanding or misinterpretation of the problem |
| Application of Concepts          | 30% | Accurately applies appropriate concepts to solve complex problems          | Applies relevant concepts correctly with minor errors | Applies basic concepts with limited accuracy   | Fails to apply appropriate concepts                    |
| Problem-Solving Approach         | 20% | Logical, well-structured, and efficient approach                           | Mostly logical approach with minor gaps               | Basic approach with some inconsistencies       | Illogical or unclear approach                          |
| Accuracy of Solution             | 20% | Completely correct solution with proper justification                      | Mostly correct with minor errors                      | Partially correct solution                     | Incorrect or incomplete solution                       |
| Clarity                          | 10% | Clear, well-organized, and properly explained solution                     | Understandable with minor clarity issues              | Limited clarity and explanation                | Poorly presented and difficult to understand           |

**Marks Evaluation:**

| Question No.               | Understanding of Problem Context (20%) | Application of Concepts (30%) | Problem-Solving Approach (20%) | Accuracy of Solution (20%) | Clarity (10%) | Total Marks |
|----------------------------|----------------------------------------|-------------------------------|--------------------------------|----------------------------|---------------|-------------|
| <b>Q1</b><br>(50 Marks)    |                                        |                               |                                |                            |               |             |
| <b>Q2</b><br>(50 Marks)    |                                        |                               |                                |                            |               |             |
| <b>Overall Total (100)</b> |                                        |                               |                                |                            |               |             |

| Faculty In-charge | Course Coordinator | Head of Department |
|-------------------|--------------------|--------------------|
| Signature: _____  | Signature: _____   | Signature: _____   |
| Date: _____       | Date: _____        | Date: _____        |

# Q1. Cybersecurity Threat Analysis System

## Problem:

A cybersecurity system analyzes threats using the recurrence:

$$T(n) = T(n - 1) + n \log n$$

Apply the substitution method, derive the complexity, analyze scalability, and interpret using asymptotic notation.

## 1. Understanding of Problem Context

- The cybersecurity system processes threat records sequentially.
- Each new threat requires processing the previous results plus additional analysis.
- The recurrence represents cumulative processing over all threat data.
- As threat volume increases, computation also increases.
- The objective is to determine the algorithm's overall time complexity.

## 2. Application of Concepts

- Applied the **Substitution Method** to expand the recurrence.
- Repeatedly substituted until reaching the base case.
- Converted the recurrence into a summation.
- Used summation properties to simplify the expression.
- Applied asymptotic notation to express the final complexity.

## 3. Problem-Solving Approach

- Write the recurrence:

$$T(n) = T(n - 1) + n \log n$$

- Expand recursively:

$$T(n) = T(n - 2) + (n - 1)\log(n - 1) + n \log n$$

- Continue expansion:

$$T(n) = T(1) + \sum_{i=2}^n i \log i$$

- Approximate the summation:

$$\sum i \log i = \Theta(n^2 \log n)$$

- Therefore,

$$T(n) = \Theta(n^2 \log n)$$

#### 4. Accuracy of Solution

- The recurrence was expanded correctly.
- The summation form was obtained accurately.
- Growth of the summation was evaluated correctly.
- Final time complexity is:

$$\boxed{\Theta(n^2 \log n)}$$

- The solution satisfies the substitution method.

#### 5. Clarity

- Each substitution step was shown clearly.
- Mathematical derivation followed a logical sequence.
- Complexity was expressed using asymptotic notation.
- Scalability analysis was explained in simple terms.
- The conclusion clearly states that execution time increases rapidly for large threat datasets.

## Q2. E-Commerce Recommendation System

### Problem:

A recommendation engine follows:

$$T(n) = 3T(n/3) + n$$

Apply the Master Theorem and determine the complexity.

### 1. Understanding of Problem Context

- The recommendation engine divides users into smaller groups.
- Each group is processed independently.
- Recommendation results are then combined.
- The recurrence models divide-and-conquer processing.
- The goal is to analyze algorithm efficiency as user data grows.

## 2. Application of Concepts

- Applied the **Master Theorem**.
- Identified the parameters:
  - $a = 3$
  - $b = 3$
  - $f(n) = n$
- Computed:

$$n^{\log_b a} = n^{\log_3 3} = n$$

- Compared  $f(n)$  with  $n^{\log_b a}$ .
- Selected the correct Master Theorem case.

## 3. Problem-Solving Approach

- Given:

$$T(n) = 3T(n/3) + n$$

- Identify:
  - $a = 3$
  - $b = 3$
  - $f(n) = n$
- Compute:

$$n^{\log_3 3} = n$$

- Since

$$f(n) = \Theta(n^{\log_b a})$$

- Apply **Master Theorem Case 2**.
- Therefore,

$$T(n) = \Theta(n \log n)$$

## 4. Accuracy of Solution

- Parameters were identified correctly.
- Correct Master Theorem case was selected.
- Mathematical comparison was accurate.
- Final complexity obtained is:

$$\boxed{\Theta(n \log n)}$$

- The solution correctly represents divide-and-conquer efficiency.

## 5. Clarity

- Inputs to the Master Theorem were clearly identified.
- Each calculation was shown step-by-step.
- Case selection was properly justified.
- Final complexity was expressed using asymptotic notation.
- The conclusion explains that the recommendation system scales efficiently even with increasing users.